Coconut Library Tool: A Web-Based Application for Advanced Textual Analysis for Librarians

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INTRODUCTION

The rapid development of technology has inevitably led to an explosion of information. What was once straightforward data has now evolved into the vast and complex landscape. This transformation requires additional effort to extract meaningful insights. In the realm of libraries, this shift is equally prevalent, demanding librarians to embrace new approaches to textual analysis, such as leveraging machine learning and artificial intelligence (ACRL, 2020).

While certain data science methods have become widely adopted in libraries to facilitate daily tasks, their application is not universally straightforward due to various factors. These include differences in library types, financial constraints, technological variations, human resource disparities, and varying levels of staff experience. In response to these challenges, we present the "Coconut Library Tool," a web-based application that utilizes cutting-edge Natural Language Processing (NLP) technologies.

AIM

This study aims to design an open application specifically intended for textual analysis use by librarians. This application boasts a user-friendly design that eliminates the need for librarians to possess advanced computer skills or install any software on their devices.

METHOD

In this tool, we have used multiple algorithmic NLP techniques at the backend, including topic modeling, lemmatization, stemming, and network analysis & visualization.

Algorithms

- ★ Topic Modeling
  - LDA (Sievert & Shirley, 2014)
  - Biterm (Yan, Guo, Lan, & Cheng, 2013)
  - BERTopic (Grootendorst, 2022)
- ★ Network Text Analysis (Santosa, 2023)
- ★ Sunburst Visualization
- ★ Text Pre-Processing (Bird, Loper, & Klein, 2009)

Deployment

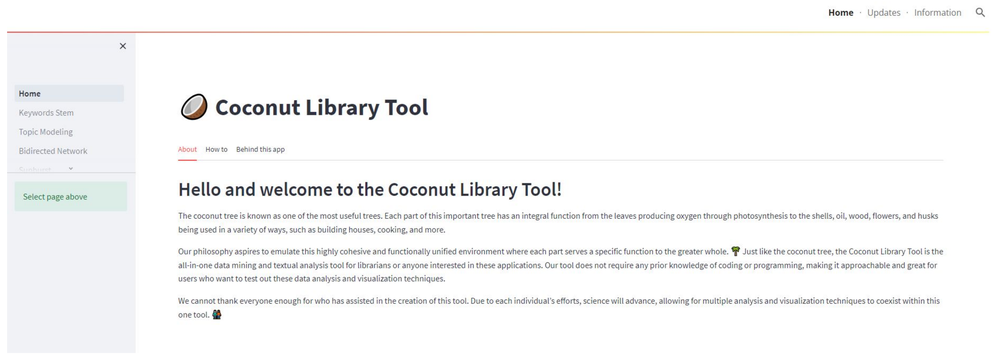
- ★ Hugging Face Space with Streamlit SDK

Data Source

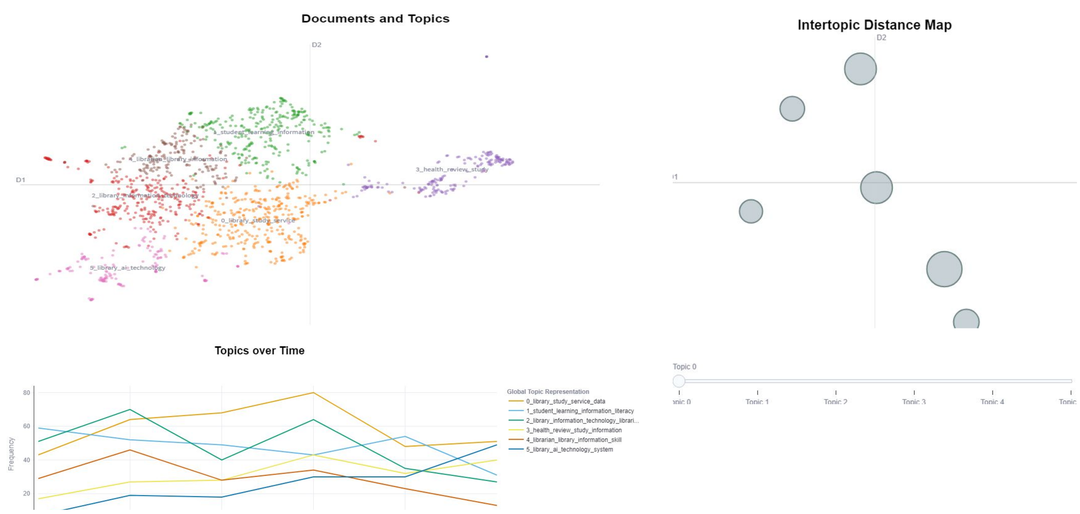
Supported file:

- ★ Scopus
- ★ Web of Science
- ★ Lens
- ★ Personalized CSV files

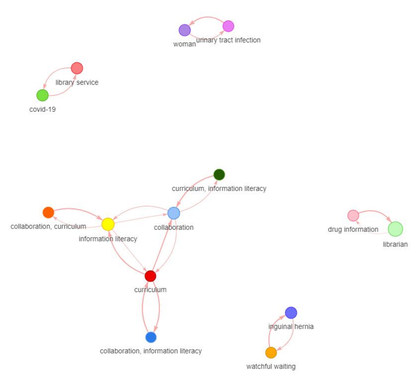
INTERFACE PREVIEW



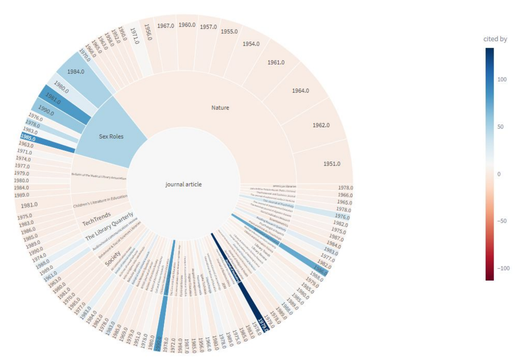
a) Coconut Library Tool interface with instructions for use in the 'How to' section.



b) Example of using BERTopic with data from Scopus. Users simply need to upload the CSV file and configure the available parameters.



c) The utilization of a bidirectional network based on the association rules mining algorithm.



d) Sunburst visualization helps to evaluate data based on type, journal title, number of documents, total citations, and average citations.

DATA AVAILABILITY STATEMENT

All original code is deposited at the Zenodo and are publicly available:
Santosa, F. A., George, C. J., & Lamba, M.. (2023). Coconut Library Tool. Zenodo. https://doi.org/10.5281/zenodo.8323458

IMPLICATIONS

Using this tool, librarians can perform (i) **topic modeling** for making ontologies, automatic subject classification, recommendation services, bibliometrics, altmetrics, and better resource searching and retrieval (Lamba & Madhusudhan, 2021); (ii) **network text analysis** to improve knowledge discovery, obtain deeper insights, and support scholars meaningfully, ultimately enhancing the library's services and resources; (iii) **sunburst visualization** to easily browse and grasp the relationships between different levels of the hierarchy; and (iv) **text pre-processing** which includes lemmatization & stemming for locating basic words and aids in catching the true meaning of the word, which can lead to improved semantic analysis and comprehension of the text.

The primary aim of this application is to directly support librarians in delivering high-quality services, promoting sustainable collaboration among diverse stakeholders, and demonstrating the inclusive and equitable use of artificial intelligence tools by librarians across the board.

WEB-APP

GITHUB

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